

Seated-Walking: A Walking-in-Place Technique for Seated VR

Liwei Chan

National Yang Ming Chiao Tung University
Hsinchu, Taiwan

liweichan@nycu.edu.tw

Zung-Hao Hsueh

National Yang Ming Chiao Tung University
Hsinchu, Taiwan

hsuehar572.cs09@nycu.edu.tw

Tzu-Wei Mi

National Yang Ming Chiao Tung University
Hsinchu, Taiwan

mitzw.cs08@nycu.edu.tw

Ming-Yun Hsu

National Yang Ming Chiao Tung University
Hsinchu, Taiwan

t311553006.cs11@nycu.edu.tw

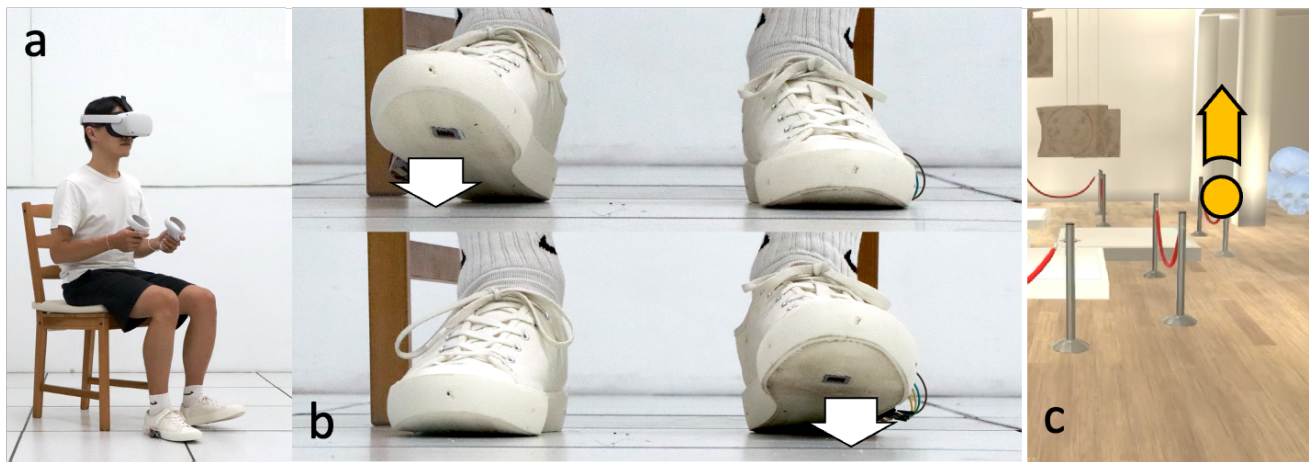


Figure 1: (a) Using Seated-Walking, the user seated in a chair, (b) takes forefoot steps (c) to advance forward in a virtual museum.

ABSTRACT

We present *Seated-Walking*, a footstep locomotion technique designed for use when sitting on a chair. *Seated-Walking* is driven by users' forefoot or rearfoot stepping, which is meant to embody the foot motions of real walking while mitigating potential fatigue. We demonstrate Seated-Walking in two applications. The first application delivers a casual virtual showroom experience, where users sitting iOptoma Corporationn a chair are allowed to walk navigating artworks in the virtual environment. The second application is an intense survival shooter experience where users track and fight enemies while avoiding their attacks, all while seated walking.

ACM Reference Format:

Liwei Chan, Tzu-Wei Mi, Zung-Hao Hsueh, and Ming-Yun Hsu. 2023. Seated-Walking: A Walking-in-Place Technique for Seated VR. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '23 Immersive Pavilion)*, August 06–10, 2023, Los Angeles, CA, USA.

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SIGGRAPH '23 Immersive Pavilion, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0151-1/23/08.

<https://doi.org/10.1145/3588027.3595603>

CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588027.3595603>

1 INTRODUCTION

One of the key design challenges to the practicality of virtual reality (VR) is to accommodate the mismatch between the user's virtual and physical experiences while preserving their natural interaction. Walking-in-place techniques [Tregillus and Folmer 2016], for example, deal with the mismatch by allowing users to navigate a virtual environment by walking in place in a constrained physical space.

We present *Seated-Walking*, a walking-in-place footstep locomotion technique designed for use when users are seated in a chair. Figure 1 shows a user making forefoot stepping to advance forward in a virtual museum. The footstep motions are measured with a pair of motion-sensing shoes made to be self-contained, in support of its use unbound to location. To address motion sickness that can occur *Seated-Walking*, we implemented a technique called "snapping" for each footstep in locomotion [Farmani and Teather 2020], which ensures an immediate transition corresponding to each step, minimizing discomfort. We propose a footstep locomotion design that maps forefoot and rearfoot steps to a range of locomotion actions.

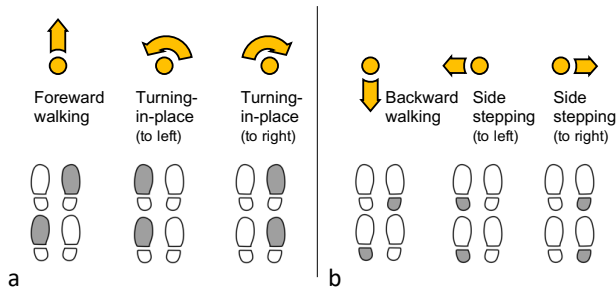


Figure 2: Footfore and rearfoot steps mapped to locomotion actions, taking the forefoot as the primary stepping method. The gray region indicates the part of the foot making the step.

2 SEATED WALKING

The design goal is to create foot commands using forefoot and rearfoot stepping in combination for activating four types of locomotion actions: forward-stepping, forwarding-while-turning, turning-in-place, side-stepping, and backward-stepping, as depicted in Figure 2. The design is made to function on non-swivel chairs maximizing the opportunity of application. We discuss the following design with forefoot-stepping as the primary stepping method and rearfoot-stepping as the secondary stepping method. In practice, they are interchangeable, and users can select whichever is preferred.

2.0.1 Alternating forefoot (primary) stepping for forward-walking. Forward-walking is activated when users alternate their forefoot steps as if walking normally. We activated the user’s virtual body to move forward by one step length in the forward direction (e.g., the body-facing) for each consecutive alternating forefoot step. When a user generally faces forward less than 15 degrees, walk-forward is activated and they move toward the exact direction that the body faces; otherwise, forwarding-while-turning is activated.

2.0.2 One-sided forefoot (primary) stepping for turning-in-place. We exploit one-sided forefoot stepping to turn the user’s virtual body to the side of the stepping foot by 22.5 degrees per consecutive same-sided forefoot stepping. This is useful (1) when users need an immediate turn at a corner and (2) when they want to remove the neck rotation imposed from using forwarding-while-turning by re-orienting their body-facing to their heading direction. The 22.5 degrees of rotation is set to gain efficiency in rotation while preserving user orientation by keeping sufficient views before and after a rotation is made.

2.0.3 Alternating rearfoot (secondary) stepping for backward-walking. Backward-walking is an infrequent but important action in correcting an exceeding motion. Here, users activate backward-stepping with alternating rearfoot stepping. Like forward-stepping, the user’s virtual body would move backward by one-step in length for each consecutive alternating rearfoot step contacting the ground.

2.0.4 One-sided rearfoot (secondary) stepping for side-stepping. We exploit one-sided rearfoot stepping to move the user’s virtual body



Figure 3: The Seated-Walking prototype including two photo-reflective sensors inserted into the fore and rear part of the shoe bottom, and WiFi Arduinos on the side.

to the side of the stepping foot by one-step length per consecutive same-sided rearfoot stepping.

3 SYSTEM IMPLEMENTATION

We aim to create an implementation that is self-contained, supporting users to use the seated-walking function whenever needed. The implementation embeds two infrared photoreflexive sensors (SG-105) at the rear and fore parts of the shoe bottom facing down, such that the fore and rear sensors would both touch the ground when the shoe is set flat on the ground (Figure 3). The sensors report the intensity of reflection per 30 microseconds, allowing us to detect users’ feet acting forefoot or rearfoot stepping in real-time. On each of the shoes, the sensors are wired through the inner of the shoe to a Wi-Fi ready Arduino (Adafruit Feather M0) linked with a lithium battery put together in a 3D-printed case fixed to the dorsal side of the shoe bottom. Sensor readings are wirelessly sent to the Unity application (ver. 2019.4.20f1) run on a VR-ready PC. (Aorus 17 ye5 laptop, Intel i7-core CPU, GeForce rtx3080ti).

4 CONCLUSION

As VR devices grow more prevalent in our daily lives, a seamless locomotion approach that allows users to effortlessly walk in VR while remaining comfortably seated in the real world will be crucial. Seated-Walking is a footstep-based walking-in-place locomotion technique designed to be used when users are seated in highly constrained spaces, such as in a car or on a plane.

ACKNOWLEDGMENTS

This research was supported in part by the National Science and Technology Council of Taiwan (MOST109-2628-E-009-010-MY3; 111-3111-E-002-002) and Optoma Corporation.

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